TWO CONSECUTIVE POWERS OF AN ELEMENT LYING IN A SUBRING PRODUCE AN ELEMENT WHOSE SQUARE AND CUBE LIE IN THE SUBRING

THE STACKS PROJECT AUTHORS

Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file pinned-original-source.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. Following the multi-claim convention for this source, source_url is the source-level URL https://stacks.math.columbia.edu/ and the source label lemma-nth-and-nplusone-implies-square-and-cube is carried in claim_key and here rather than in a per-claim tag URL.

Editorial note. The preamble is the chapter's shared Stacks Project preamble, retained as sources/stacks-morphisms/retained-preamble.tex: a line-for-line excerpt of the shared Stacks preamble staged with this snapshot. No macro was invented or redefined, and this card adds no preamble text of its own.

Editorial note. Statement span 11596-11603 and proof span 11605-11612 come from the halfA pool card operations/pool-cards/stacks-morphisms-halfA.json and were re-read in the pinned chapter file before this card was written.

Editorial note. The proof writes the exponent identity (b^n)^k (b^{n+1})^i = b^{(k+i)n+i} with the author's own notation; it is preserved.

Allowed standard prerequisites (editorial).

- Arithmetic of powers in a ring and the description of the powers of an element as a numerical semigroup generated by two consecutive integers. Used at proof lines 11822-11825, where the identity expressing b to the power (k+i)n+i is written down. Shows that all sufficiently large powers of the given element lie in the subring, with the explicit bound n squared.
- Existence of a largest exponent with the property that the corresponding power does not lie in the subring, using that all large powers do lie in it. Used at proof lines 11826-11828. Produces the element whose square and cube are in the subring.

Lemma 0.1. *Let $A \subset B$ be a ring extension. If there exists $b \in B$, $b \notin A$ and an integer $n \geq 2$ with $b^n \in A$ and $b^{n+1} \in A$, then there exists a $b' \in B$, $b' \notin A$ with $(b')^2 \in A$ and $(b')^3 \in A$.*

Proof. Let b and n be as in the lemma. Then all sufficiently large powers of b are in A. Namely, $(b^n)^k(b^{n+1})^i = b^{(k+i)n+i}$ which implies any power b^m with $m \geq n^2$ is in A. Hence if $i \geq 1$ is the largest integer such that $b^i \notin A$, then $(b^i)^2 \in A$ and $(b^i)^3 \in A$. □

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